

Generation Interconnection System Impact Study Report

For Queue Project Q17-0808

Parke 138 kV

May 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0808. The Interconnection Customer proposes a 102.3 MW battery energy storage facility interconnecting at the Parke 138 kV substation in the Virginia area. The request was received on June 09, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 371 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 48 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$35,159,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Coyote to Hobrt 138kV	Clinch Riv to Carton Hills 345kV line	166.2	219	7.5
S Windsor to Dexter Tap 138kV	Kean Mtn And Grassy Creek to Barr Lake 345kV	156.5	149	6.3
Snarr to Rhinelander Paper 138kV	Parke to Barr Lake 345kV Line	154.5	203	55.2
Dewey to Tillamook 138kV	Davisville to N Appleton 138kV	154.0	191	5.9
Glen Lyn to Snarr Substation 138kV line	Parke to Barr Lake 345kV L	151.7	213	55.2
Tillamook to Pocket 138kV	N Appleton to Davisville Substation 138kV	149.8	191	5.9
Heyburn to Lysander 138kV Line	Sidney Term to Moadore 138kV Line	147.8	191	5.8
Dewey to Dowell 138kV	North Appleton to Davisville 138kV L	144.0	210	5.9
Rochester to Radford Army Ammunition Plant 138kV line	Hinton Ap to Lowmoor 345kV	142.3	183	15.1
Tilton to South Windsor 138kV line	Kean Mtn And Grassy Crk to Barr Lake 345kV	141.2	166	6.3
Mt Vernon to Pocket 138kV	N Appleton to Davisville 138kV	136.2	203	5.9
Buckley to Celanese Acetate Llc 138kV	Lowmoor to Hinton Ap 345kV line	135.1	198	10.1
Lysander to Raver Substation 138kV	Sidney Term to Mogadore 138kV L	133.9	217	5.8
Rochester to Sysquoc 138kV Line	Lowmoor to Hinton Ap 345kV line	132.3	209	15.1
Raver to Sherman Road 138kV	Sidney Term to Mogadore 138kV L	132.3	222	5.8
Mount View to Peyton 138kV	Lowmoor to Hinton Ap Substation 345kV	132.2	167	10.1
Rhinelander Paper to Ft White 138kV line	Hinton Ap to Lowmoor 345kV	131.6	219	10.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Buckley to Ft White 138kV	Lowmoor to Hinton Ap 345kV	131.0	212	10.1
Mcclain Energy Facility to Coal Substation 138kV line	Manatee to Hinton Ap 345kV	130.6	254	10.6
Devilliers to Crater 138kV	Meadow Bridge to Antelope Vly Station 138kV	130.5	167	5.6
Devilliers to Batesville 138kV	Meadow Bridge to Antelope Valley Sta 138kV	130.5	164	5.6
Taloga to Pomona Hts 138kV L	Lowmoor to Hnton Ap 345kV	129.3	252	15.1
Winnebago Wind Power Project to Clinch Riv 138kV	Barr Lake to Kea Mtn And Grassy Creek 345kV	128.7	246	5.9
Pringle to S. Land Park & 35Th Ave 138kV	Manatee to New Hope Ap 345kV	128.4	216	8.3
Orovada to Carlton Hills 138kV	Clinch River to Carlton Hills 345kV line	127.9	259	5.6
Ingles St to Svp 138kV	Hinton Ap to Lowmoor 345kV	127.4	191	15.1
Haysi to Leesburg Picciola 138kV	Carlton Hills to Atoka 345kV	126.8	262	10.5
Winnebago Wind Power Project to Lebanon 138kV	Kean Mtn And Grassy Creek to Barr Lake 345kV	126.2	242	5.9
North Claytor to Svp 138kV line	Hinton Ap to Lowmoor 345kV	126.2	193	15.1
Byhalia to Lebanon 138kV line	Barr Lake to Kean Mtn And Grassy Crk 345kV	124.8	235	5.9
N Claytor to Radford Army Ammunition Plant 138kV Line	Lowmoor to Hinton Ap 345kV	124.6	211	15.1
Spring Crk Road to Harris 138kV line	Sidney Coal to Meeker 138kV	123.7	315	6.7
Turkey Crk to Meeker Substation 138kV	Spring Crk Rd to Harris 138kV	123.4	219	5.8
Crater to Heyburn 138kV	Meadow Bridge to Antelope Valley Sta 138kV	123.1	181	5.6
Byhalia to Calpeak Pe 7 Substation 138kV	Barr Lake to Kean Mtn And Grassy Creek 345kV	123.1	231	5.9
Berlin Ctr to Dexter Tap 138kV	Kean Mtn And Grassy Crk to Barr Lake 345kV	123.1	183	6.3
Sidney Coal to Meeker 138kV	Spring Creek Road to Harris 138kV Line	122.8	231	5.8
S Vicksburg to Celanese Acetate Llc 138kV Line	Hinton Ap to Lowmoor 345kV	121.7	222	10.1
Kimballton to Covadonga 1014-1013-1012-1011 138kV	Lowmoor to Hinton Ap 345kV	121.1	218	10.1
S Vicksburg to Kimballton 138kV	Lowmoor to Hinton Ap 345kV	120.8	219	10.1
Covadonga 1014-1013-1012-1011 to Peyton 138kV Line	Lowmoor to Hinton Ap 345kV	119.8	212	10.1
Lake Forrest to Sysquoc 138kV line	Lowmoor to Hinton Ap 345kV	119.7	243	15.1
Beardsley St to Red Hills 138kV	Lowmoor to Hinton Ap 345kV	118.6	355	10.4
Lake orrest to Lytle 138kV	Lowmoor to Hinton Ap 345kV	118.1	256	15.1
Mount View to Hartman Crk Substation 138kV	Lowmoor Substation to Hiton Ap 345kV	117.9	217	10.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pomona Hts to Lytle 138kV	Lowmoor to Hinton Ap 345kV	116.5	270	15.1
Pringle to Halls Ridge 138kV	New Hope Ap to Mantee 345kV	114.8	243	8.3
Hinton Ap 345 kV/138 kV XFMR	Manatee to Hinton Ap 345kV Line	111.4	640	11.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Coyote to Hobrt 138kV	Rebuild line with higher-capacity conductor	\$5,171,000	5% (\$252,000)
Dexter Tap to S Windsor 138kV L	Rebuild line with higher-capacity conductor	\$8,665,000	23% (\$2,023,000)
Snarr to Rhinelander Paper 138kV	Rebuild line with higher-capacity conductor	\$5,898,000	44% (\$2,614,000)
Dewey to Tillamok 138kV	Rebuild line with higher-capacity conductor	\$7,363,000	4% (\$275,000)
Glen Lyn to Snarr Substation 138kV line	Rebuild line with higher-capacity conductor	\$3,046,000	44% (\$1,350,000)
Tillamook to Pocket 138kV	Rebuild line with higher-capacity conductor	\$5,333,000	4% (\$199,000)
Heyburn to Lysander 138kV	Rebuild line with higher-capacity conductor	\$12,372,000	4% (\$519,000)
Dewey to Dowell 138kV	Rebuild line with higher-capacity conductor	\$8,159,000	4% (\$305,000)
Rochester to Radford Army Ammunition Plant 138kV line	Rebuild line with higher-capacity conductor	\$9,617,000	11% (\$1,020,000)
Tilton to South Windsor 138kV line	Rebuild line with higher-capacity conductor	\$23,489,000	23% (\$5,485,000)
Pocket Substation to Mt Vernon 138kV	Rebuild line with higher-capacity conductor	\$7,453,000	4% (\$279,000)
Buckley to Celanese Acetate Llc 138kV	Rebuild line with higher-capacity conductor	\$6,938,000	12% (\$819,000)
Lysander to Raver 138kV	Rebuild line with higher-capacity conductor	\$5,117,000	4% (\$215,000)
Rochester to Sysquoc 138kV	Rebuild line with higher-capacity conductor	\$5,563,000	11% (\$590,000)
Raver to Sherman Road 138kV	Rebuild line with higher-capacity conductor	\$4,012,000	4% (\$168,000)
Peyton to Mount View 138kV	Rebuild line with higher-capacity conductor	\$6,414,000	12% (\$757,000)
Rhinelander Paper to Ft White 138kV line	Rebuild line with higher-capacity conductor	\$10,376,000	12% (\$1,225,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Buckley to Ft White 138kV Line	Rebuild line with higher-capacity conductor	\$4,722,000	12% (\$558,000)
Mcclain Energy Facility to Coal Substation 138kV line	Rebuild line with higher-capacity conductor	\$5,690,000	6% (\$328,000)
Devilliers to Crater 138kV	Rebuild line with higher-capacity conductor	\$10,295,000	4% (\$435,000)
Devilliers to Batesville 138kV	Rebuild line with higher-capacity conductor	\$9,618,000	4% (\$407,000)
Taloga to Pomona Hts 138kV L	Rebuild line with higher-capacity conductor	\$21,368,000	11% (\$2,266,000)
Winnebago Wind Power Project to Clinch Riv 138kV	Rebuild line with higher-capacity conductor	\$7,303,000	7% (\$481,000)
Pringle to S. Land Park & 35Th Ave 138kV	Rebuild line with higher-capacity conductor	\$3,893,000	6% (\$225,000)
Orovada to Carlton Hills 138kV	Rebuild line with higher-capacity conductor	\$845,000	7% (\$62,000)
Ingles St to Svp 138kV	Rebuild line with higher-capacity conductor	\$1,395,000	11% (\$148,000)
Haysi to Leesburg Picciola 138kV line	Rebuild line with higher-capacity conductor	\$14,762,000	8% (\$1,222,000)
Winnebago Wind Power Project to Lebanon 138kV	Rebuild line with higher-capacity conductor	\$3,171,000	7% (\$209,000)
N Claytor to Svp 138kV	Rebuild line with higher-capacity conductor	\$7,345,000	11% (\$779,000)
Byhalia to Lebanon 138kV	Rebuild line with higher-capacity conductor	\$10,390,000	7% (\$685,000)
N Claytor to Radford Army Ammunition Plant 138kV Line	Rebuild line with higher-capacity conductor	\$2,336,000	11% (\$248,000)
Spring Crk Rd to Harris 138kV	Rebuild line with higher-capacity conductor	\$4,457,000	3% (\$115,000)
Turkey Crk to Meeker Substation 138kV	Rebuild line with higher-capacity conductor	\$710,000	3% (\$21,000)
Crater to Heyburn 138kV	Rebuild line with higher-capacity conductor	\$1,805,000	4% (\$76,000)
Byhalia to Calpeak Pe 7 138kV L	Rebuild line with higher-capacity conductor	\$19,216,000	7% (\$1,267,000)
Berlin Ctr to Dexter Tap 138kV	Rebuild line with higher-capacity conductor	\$6,986,000	23% (\$1,631,000)
Sidney Coal to Meeker 138kV	Rebuild line with higher-capacity conductor	\$5,857,000	3% (\$177,000)
S Vicksburg to Celanese Acetate Llc 138kV Line	Rebuild line with higher-capacity conductor	\$5,602,000	12% (\$661,000)
Kimballton to Covadonga 1014-1013-1012-1011 138kV	Rebuild line with higher-capacity conductor	\$7,360,000	12% (\$868,000)
S Vicksburg to Kimballton 138kV	Rebuild line with higher-capacity conductor	\$699,000	12% (\$82,000)
Covadonga 1014-1013-1012-1011 to Peyton 138kV Line	Rebuild line with higher-capacity conductor	\$12,416,000	12% (\$1,465,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lake Forrest to Sysquoc 138kV line	Rebuild line with higher-capacity conductor	\$4,584,000	11% (\$486,000)
Beardsley St to Red Hills 138kV	Rebuild line with higher-capacity conductor	\$16,976,000	2% (\$423,000)
Lake Forrest to Lytle 138kV	Rebuild line with higher-capacity conductor	\$2,825,000	11% (\$300,000)
Mount View to Hartman Crk Substation 138kV	Rebuild line with higher-capacity conductor	\$2,845,000	8% (\$226,000)
Pomona Hts to Lytle 138kV	Rebuild line with higher-capacity conductor	\$3,892,000	11% (\$413,000)
Pringle to Halls Ridge 138kV	Rebuild line with higher-capacity conductor	\$8,906,000	6% (\$515,000)
Hinton Ap 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$15,812,000	2% (\$285,000)
	Total allocated to project		\$35,159,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.